Uisang Lee

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EDUCATION

Seoul National University Sep 2021 – Aug 2027 (Expected)
Integrated M.S./Ph.D. Program in Electrical and Computer Engineering
Advisor: Prof. Soo-Mook Moon · GPA: 3.62 / 4.5

Kyung Hee University Mar 2015 – Aug 2021
B.S. in Biomedical Engineering
GPA: 3.43 / 4.5

RESEARCH EXPERIENCE

VMO Laboratory, Seoul National University — Graduate Researcher Sep 2021 – Present
Conducted vulnerability analysis of Solidity-based blockchain code and developed a Graph Neural Network (GNN)–based vulnerability detection method, resulting in a first-author paper selected for an Oral presentation at AAAI 2026.

Currently researching LLM-based code generation and response-generation methods that accurately reflect user prompt intent.

EXPERIENCE

VMO Laboratory, Seoul National University — Graduate Researcher Sep 2021 – Present
Implemented IR code and performed pattern analysis for an ATE Pattern Program through an industry–academia project with Samsung Electronics DS Memory Business Unit.

Participated in an ETRI blockchain analysis project, producing four domestic papers, one paper at a top-tier AI conference, and one patent.

Served as a teaching assistant for the Introduction to Compilers and Advanced Compilers courses at Seoul National University.

TEACHING EXPERIENCE

Seoul National University — Teaching Assistant, Introduction to Compilers Sep 2022 – Dec 2022
Graded assignments and exams for a course covering compiler fundamentals, including syntax processing of programming languages (Dept. of Electrical and Computer Engineering, Fall 2022).

Seoul National University — Teaching Assistant, Advanced Compilers Mar 2023 – Jun 2023
Graded assignments and exams for a course on compiler back-end optimization techniques (Dept. of Electrical and Computer Engineering, Spring 2023).

Samsung Electronics DS — Instructor, System Expert Program Apr 2023 – May 2024
Delivered lectures and conducted evaluations on compiler front-end and back-end topics (Introductory and Advanced Compilers) for Samsung Electronics DS employees.

HONORS & AWARDS

Hyundai Motor Chung Mong-Koo Foundation Aug 2023 – Jun 2026
Chung Mong-Koo Scholarship — Science & Technology Scholarship (Graduate)

POSTECH Open Innovation Big Data Center Oct 2020
Encouragement Award, 2nd POSTECH Open Innovation Big Data Challenge

PUBLICATIONS

BugSweeper: Function-Level Detection of Smart Contract Vulnerabilities Using Graph Neural Networks 2026
Proceedings of the AAAI Conference on Artificial Intelligence, 40(1), 570–578 (Oral)
Uisang Lee (first author), C. Chung, J. Lee, S.-M. Moon

CONFERENCE PRESENTATIONS

AAAI 2026, Singapore — Oral Presentation 2026
BugSweeper: Function-Level Detection of Smart Contract Vulnerabilities Using Graph Neural Networks

RESEARCH INTERESTS

Response-generation methods that enable LLMs — including diffusion LLMs (dLLMs) and autoregressive (AR) models — to accurately understand and reflect prompt intent.

Training and inference methods for improving the generation accuracy and reasoning capability of LLMs in the code domain.

TECHNICAL SKILLS

Python, C/C++, PyTorch

LANGUAGES

English: Fluent
Japanese: Intermediate